

Model Development Document: Mortgage 24-Month Probability of Default (v1.1)

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Numbers are generated by `src/build_report.py` from the pipeline outputs.

1. Intended use

Rank-order and estimate the probability that a newly originated, Freddie Mac-acquired single-family fixed-rate mortgage reaches default within 24 months of origination, using only information known at origination. Illustrative scope: application scoring and portfolio PD for a prime mortgage book. **Not** intended for subprime or non-conforming loans, adjustable-rate loans, or lifetime/IFRS 9 staging without further work. This is a portfolio project, not a production model.

2. Data and population

- Source: Freddie Mac Single Family Loan-Level Dataset, **sample files** (50,000 random loans per origination year), July 2026 file layout. Freddie Mac drew the samples; no further sampling was applied. 699,999 loans are used (one stray loan with a 2009 origination was dropped).
- Years used: 1999-2008 and 2016-2019 (14 vintages).
- Population: acquired (accepted) loans only. Rejected applicants are not observed, so there is no reject inference.

Sample	Origination years	Loans	Bad rate	Use
Development train	1999-2004 (70%)	209,866	1.19%	fit both models
Development hold-out	1999-2004 (30%)	90,134	1.26%	in-time test
Out-of-time	2005	50,000	2.12%	stability
Crisis	2006-2008	149,999	4.68%	stress
Recent regime	2016-2019	200,000	0.71%	different environment

The 70/30 split is random with seed 42, applied after sorting loans by identifier so it is reproducible.

3. Target definition

Bad = within 24 months of origination (loan age at most 24), the loan is 90+ days delinquent, in REO acquisition (status RA), or leaves the book with zero-balance code 02 (third-party sale), 03 (short sale or charge-off), 09 (REO disposition) or 15 (whole loan sale). Codes 01 (payoff), 16 and 96 are not defaults. Prepaid loans count as good.

COVID-19 adjustment. Delinquencies are not counted when the loan is in forbearance (Borrower Assistance Plan = F) or flagged disaster-related. Without it, 83% of 2018 and 98% of 2019 "bad" loans first reached 90+ days from March 2020, giving bad rates of 2.7% and 4.6%. With it they are 0.74% and 0.66%.

Limitations. (a) Censoring: loans that prepay cannot default later, understating 24-month risk for fast-prepaying vintages; a discrete-time hazard model is a possible extension. (b) Code 15 inclusion is a

judgement call. (c) A forbearance borrower who later loses the home is still counted through the sale codes.

4. Features and leakage control

Origination-file fields only: credit score, LTV, CLTV, DTI, original balance, interest rate, MI %, units, borrowers, first-time buyer, occupancy, channel, loan purpose, property type, state. Nothing from the performance files is a feature (enforced by `tests/test_leakage.py`).

Placeholder codes (credit score 9999; LTV, CLTV, DTI, MI % 999; units, borrowers, property type 99; "9" on categoricals) are set to missing. Missing credit score (0.3% of loans) and DTI (2.4%) get a missing-indicator; in the scorecard missing is its own bin.

5. Champion: WoE logistic scorecard

- Binning on train only: quantile start, bins merged to at least 5% of loans, monotone bad rate enforced for credit score (down), and for LTV, CLTV, DTI and rate (up).
- Selection: information value of at least 0.02; CLTV dropped as a near-duplicate of LTV; wrong-sign coefficients removed iteratively.
- Retained (10): credit score (IV 1.27), LTV (0.41), MI % (0.26), interest rate (0.23), borrowers (0.19), DTI (0.14), state (0.13), channel (0.09), balance (0.04), property type (0.04). Dropped for low IV: loan purpose, occupancy, first-time buyer, units.
- Logistic regression (statsmodels); all coefficients have the expected sign.
- Points: 600 at 50:1 good:bad odds, 20 points to double the odds.

6. Challenger: LightGBM

Same features, native categoricals, missing left as NaN. Tuned over a 4-setting grid with expanding-window time folds inside 1999-2004 (train on earlier years, test on the next). The shallowest model won (8 leaves, 300 trees, learning rate 0.05; mean CV AUC 0.842). Row and column subsampling and an L2 penalty limit overfitting.

7. Development performance

Model	Train AUC	Hold-out AUC	Hold-out Gini	Hold-out KS
Scorecard	0.839	0.840	0.680	0.537
LightGBM	0.880	0.849	0.699	0.550

LightGBM shows a larger train to hold-out gap (0.031 against -0.001), so it is more prone to overfit. Calibration, stress and out-of-time results are left to the independent validation.

8. Assumptions and known limitations

1. Relationships learned in 1999-2004 (a benign period) hold in other regimes; the crisis test challenges this.
2. Sample files only; about 2,505 bad loans in training, thin for tree models.
3. Accepted-only population, no reject inference.
4. State is a feature and may proxy for geography-related risk; it needs fairness review.

5. Some bins are tiny (smallest retained: `mi_pct = MISSING`, 10 loans), so estimates there are unreliable.
6. No macroeconomic variables, so the models cannot respond to the cycle.
7. Credit score has unusually high information value (1.27); treat it as a concentration risk.

9. Version history and freeze

Both models are hashed in `FROZEN.json`. Any change after validation results is logged as a new model version.

Current: v1.1. v1.0 was not reproducible: DuckDB row order changed the seeded 70/30 split between runs. Fixed by sorting on `loan_identifier` before the split. Models refit unchanged otherwise. Done for reproducibility, not performance; v1.0 model files could not be recovered.